ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

ENGINEERING PHYSICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 1st year Engineering Physics paper 1 · practice a.

Written by Academic Central for practice, following first-year Engineering Physics syllabuses at South African universities (mechanics, thermal physics and waves for engineering students). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Deduct one mark only once per question for missing or wrong units. Accept answers that differ only through rounding.

QUESTION 1

1.1 Force: N = kg·m/s² (kg·m·s⁻²) ✓ Energy: J = kg·m²/s² (kg·m²·s⁻²) ✓ (2)

1.2 Left side: [m]. Right side: ut → (m/s)(s) = m and at² → (m/s²)(s²) = m ✓ (2)
Every term has units of metres, so the equation is dimensionally correct ✓

1.3 $R = \sqrt{(30^2 + 40^2)} = 50$ N ✓ (3)
$\theta = \tan^{-1}(40/30) = 53.13°$ ✓ north of east ✓

1.4 72 ÷ 3.6 = 20 m/s ✓ (1)

4.56 × 10^{-4} m ✓
3 significant figures ✓

(2)

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QUESTION 2

2.1.1 $v = u + at = 0 + 2.5 \times 8$ ✓ = 20 m/s ✓ (2)

2.1.2 $s = ut + \frac{1}{2}at^2 = \frac{1}{2} \times 2.5 \times 8^2$ ✓ = 80 m ✓ (2)

2.2.1 $0 = 19.6 - 9.81t$ ✓ t = 2.00 s ✓ (2)

2.2.2 $h = u^2 / 2g = 19.6^2 / (2 \times 9.81)$ ✓ = 19.58 m ✓ (2)

2.2.3 2 × 2.00 = 4.00 s ✓ (1)

2.3.1 $u_x = 25 \cos 30° = 21.65$ m/s; $u_y = 25 \sin 30° = 12.5$ m/s ✓ (3)
$t = 2u_y / g$ ✓ = 2 × 12.5 / 9.81 = 2.55 s ✓

2.3.2 $R = u_x t = 21.65 \times 2.548$ ✓ = 55.17 m ✓ (2)

2.3.3 $H = u_y^2 / 2g = 12.5^2 / 19.62$ ✓ = 7.96 m ✓ (2)

2.4 Gradient = acceleration ✓ Area under the graph = displacement ✓ (2)

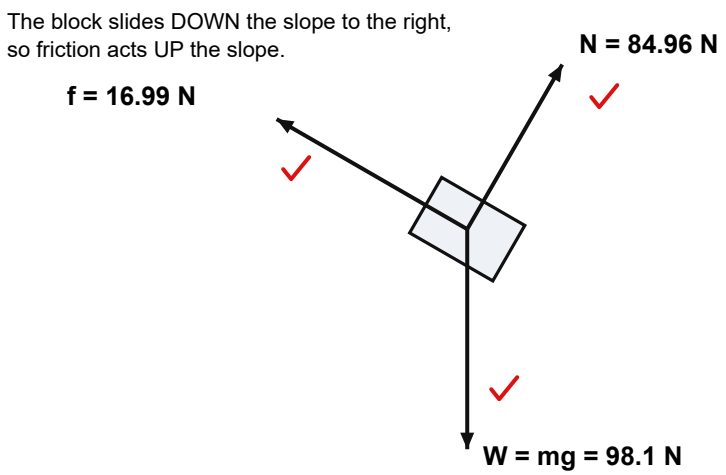
2.5 $a = v^2/r = 4^2 / 0.8 = 20$ m/s² towards the centre ✓ (2)
$F = ma = 0.5 \times 20 = 10$ N ✓

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QUESTION 3

3.1 The resultant force on a body equals the rate of change of its momentum; for a constant mass, the acceleration is directly proportional to the resultant force ✓ and inversely proportional to the mass (F = ma) ✓ (2)

3.2.1



Weight straight down ✓; normal force at right angles to the slope ✓; friction up the slope ✓ (3)

3.2.2 N = mg cos 30° = 10 × 9.81 × 0.866 ✓ = 84.96 N ✓ (2)

3.2.3 f = μN = 0.2 × 84.96 = 16.99 N ✓ (1)

3.2.4 Fnet = mg sin 30° − f = 49.05 − 16.99 ✓ = 32.06 N ✓ (3)
a = 32.06 / 10 = 3.21 m/s² down the slope ✓

3.3 T − mg = ma ✓ (3)
T = 800(9.81 + 1.5) ✓ = 9 048 N ✓

3.4.1 (5 − 3)g = (5 + 3)a ✓✓ a = 2 × 9.81 / 8 = 2.45 m/s² ✓ (3)

3.4.2 For the 3 kg mass: T − 3g = 3a ✓ (3)
T = 3(9.81 + 2.45) ✓ = 36.79 N ✓

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QUESTION 4

4.1 Work is done when a force moves its point of application in the direction of the force: W = F·s·cos θ ✓. Unit: joule (J) ✓ (2)

4.2.1 E_k = ½mv² = ½ × 1 200 × 20² ✓ = 240 000 J ✓ (2)

4.2.2 W = F × s, so F = 240 000 / 50 ✓ = 4 800 N ✓ (2)

4.3.1 W = mgh = 500 × 9.81 × 12 ✓ = 58 860 J ✓ (2)

4.3.2 P = 58 860 / 60 = 981 W ✓ (1)

4.3.3 P_{in} = 981 / 0.75 ✓ = 1 308 W ✓ (2)

4.4.1 Momentum before = momentum after ✓ (3)
2 × 3 = (2 + 1)v ✓ v = 2 m/s ✓

4.4.2 Before: ½ × 2 × 3² = 9 J; after: ½ × 3 × 2² = 6 J ✓ (3)
Loss = 3 J ✓ Kinetic energy is not conserved, so the collision is inelastic ✓

4.5 Δp = m(v − u) = 0.15(15 − (−20)) ✓ = 5.25 kg·m/s away from the wall ✓ (3)
F = Δp / Δt = 5.25 / 0.01 = 525 N ✓

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QUESTION 5

5.1 Beam weight = 196.2 N at 2 m; load = 490.5 N at 1 m from A ✓ (5)
Moments about A: $R_B \times 4 = 196.2 \times 2 + 490.5 \times 1$ ✓
$R_B = 882.9 / 4 = 220.73$ N ✓
Upward forces = downward forces: $R_A + R_B = 686.7$ N ✓
$R_A = 465.98$ N ✓

5.2 $\tau = F \times r = 40 \times 0.3 = 12$ N·m ✓ (1)

5.3 The resultant force is zero ✓ (2)
The resultant moment (torque) about any point is zero ✓

5.4.1 $\alpha = 100 / 10 = 10$ rad/s² ✓ (1)

5.4.2 $\tau = I\alpha = 2 \times 10$ ✓ $= 20$ N·m ✓ (2)

5.4.3 $E_k = \frac{1}{2}I\omega^2 = \frac{1}{2} \times 2 \times 100^2$ ✓ $= 10\,000$ J ✓ (2)

5.4.4 $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2} \times 10 \times 10^2$ ✓ $= 500$ rad ✓ (2)

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QUESTION 6

6.1 $Q = mc\Delta T = 2 \times 4\,186 \times 60$ ✓ $= 502\,320$ J (502.32 kJ) ✓ (2)

6.2 $\Delta U = Q - W$ ✓ $= 500 - 200 = 300$ J (internal energy increases) ✓ (2)

6.3 $P_1V_1/T_1 = P_2V_2/T_2$ ✓ (3)
$P_2 = 100 \times 0.02 \times 450 / (0.005 \times 300)$ ✓ $= 600$ kPa ✓

6.4 $v = f\lambda = 500 \times 0.68$ ✓ $= 340$ m/s ✓ $T = 1/f = 0.002$ s (2 ms) ✓ (3)

6.5 $\omega = \sqrt{(k/m)} = \sqrt{(200/0.5)} = 20$ rad/s ✓ (3)
$f = \omega / 2\pi = 3.18$ Hz ✓ $T = 1/f = 0.31$ s ✓

6.6 Resonance: a system vibrates with a very large amplitude when it is driven at its natural frequency ✓ (2)
Example (any one): a bridge or building in wind or earthquakes; a machine or motor mount running at the frame's natural frequency; a robot arm vibrating at its natural frequency ✓

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TOTAL: 100

END OF MARKING GUIDELINE